

EDEXCEL INTERNATIONAL A LEVEL

WMA11/01 October 2022 Worked Solutions

Oct 2022 paper rebuilt with the worked solution after each question.

9

paper questions

WMA11

Pure 1

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**

Prepared for Dr Eslam Ahmed - eliteigcse.com

P1

PAST PAPER

WMA11/01 October 2022

Oct 2022 | 9 questions | 75 marks

9

questions

75

marks

Answers

worked solution
after each
question

Question 1

Differentiation

1. The curve C has equation

$$y = \frac{x^3}{4} - x^2 + \frac{17}{x} \quad x > 0$$

- (a) Find $\frac{dy}{dx}$, giving your answer in simplest form.

(3)

The point $R\left(2, \frac{13}{2}\right)$ lies on C .

- (b) Find the equation of the tangent to C at the point R . Write your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

(3)

(Total for Question 1 is 6 marks)

Worked Solution - Question 1

1. Rewrite the curve

$$y = \frac{x^3}{4} - x^2 + \frac{17}{x} = \frac{1}{4}x^3 - x^2 + 17x^{-1}.$$

2. Differentiate

$$\frac{dy}{dx} = \frac{3}{4}x^2 - 2x - 17x^{-2} = \frac{3}{4}x^2 - 2x - \frac{17}{x^2}.$$

3. Find the gradient at R

At $x = 2$, the tangent gradient is $\frac{3}{4}(4) - 2(2) - \frac{17}{4} = 3 - 4 - \frac{17}{4} = -\frac{21}{4}.$

4. Use the point R

The point is $R\left(2, \frac{13}{2}\right)$, so $y - \frac{13}{2} = -\frac{21}{4}(x - 2).$

5. Write in integer form

Multiplying by 4 gives $4y - 26 = -21x + 42$, hence $21x + 4y - 68 = 0.$

Final answer

(a) $\frac{dy}{dx} = \frac{3}{4}x^2 - 2x - \frac{17}{x^2}.$

(b) $21x + 4y - 68 = 0.$

Question 2

Integration

2. Given that

$$(x - 5)(2x + 1)(x + 3) \equiv ax^3 + bx^2 - 32x - 15$$

where a and b are constants,

(a) find the value of a and the value of b .

(2)

(b) Hence find

$$\int \frac{(x - 5)(2x + 1)(x + 3)}{5\sqrt{x}} dx$$

writing each term in simplest form.

(5)

(Total for Question 2 is 7 marks)

Worked Solution - Question 2

1. Expand the identity

$$(2x + 1)(x + 3) = 2x^2 + 7x + 3, \text{ so}$$

$$(x - 5)(2x^2 + 7x + 3) = 2x^3 - 3x^2 - 32x - 15.$$

2. Read off a and b

Comparing with $ax^3 + bx^2 - 32x - 15$ gives $a = 2$ and $b = -3$.

3. Rewrite the integrand

$$\frac{(x - 5)(2x + 1)(x + 3)}{5\sqrt{x}} = \frac{1}{5}(2x^{5/2} - 3x^{3/2} - 32x^{1/2} - 15x^{-1/2}).$$

4. Integrate term by term

$$\int \frac{1}{5}(2x^{5/2})dx = \frac{4}{35}x^{7/2} \text{ and } \int \frac{1}{5}(-3x^{3/2})dx = -\frac{6}{25}x^{5/2}.$$

5. Complete the integral

$$\int \frac{1}{5}(-32x^{1/2})dx = -\frac{64}{15}x^{3/2} \text{ and } \int \frac{1}{5}(-15x^{-1/2})dx = -6x^{1/2}, \text{ then add } c.$$

Final answer

$$(a) a = 2, b = -3.$$

$$(b) \frac{4}{35}x^{7/2} - \frac{6}{25}x^{5/2} - \frac{64}{15}x^{3/2} - 6\sqrt{x} + c.$$

Question 3

Straight Line

3. The share price of a company is monitored.

Exactly 3 years after monitoring began, the share price was £1.05

Exactly 5 years after monitoring began, the share price was £1.65

The share price, £ V , of the company is modelled by the equation

$$V = pt + q$$

where t is the number of years after monitoring began and p and q are constants.

- (a) Find the value of p and the value of q .

(3)

Exactly T years after monitoring began, the share price was £2.50

- (b) Find the value of T , according to the model, giving your answer to one decimal place.

(2)

(Total for Question 3 is 5 marks)

Worked Solution - Question 3

1. Use the two data points

The model $V = pt + q$ passes through $(3, 1.05)$ and $(5, 1.65)$.

2. Find the gradient

$$p = \frac{1.65 - 1.05}{5 - 3} = \frac{0.60}{2} = 0.30.$$

3. Find the intercept

Using $1.05 = 0.30(3) + q$, we get $q = 0.15$.

4. Find T

For $V = 2.50$, $2.50 = 0.30T + 0.15$, so $T = \frac{2.35}{0.30} = 7.833 \dots$

5. Round correctly

To one decimal place, $T = 7.8$.

Final answer

(a) $p = 0.3$, $q = 0.15$.

(b) $T = 7.8$ years.

Question 4

Indices & Surds

4.

In this question you must show detailed reasoning.
Solutions relying on calculator technology are not acceptable.

$$f(x) = x^2(2x + 1) - 15x$$

(a) Solve

$$f(x) = 0 \quad (4)$$

(b) Hence solve

$$y^{\frac{4}{3}}\left(2y^{\frac{2}{3}} + 1\right) - 15y^{\frac{2}{3}} = 0 \quad y > 0$$

giving your answer in simplified surd form. (2)

(Total for Question 4 is 6 marks)

Worked Solution - Question 4

1. Factor $f(x)$

$$f(x) = x^2(2x + 1) - 15x = x(2x^2 + x - 15).$$

2. Factor the quadratic

$$2x^2 + x - 15 = (2x - 5)(x + 3).$$

3. Solve $f(x) = 0$

$$x(2x - 5)(x + 3) = 0, \text{ so } x = 0, x = \frac{5}{2}, \text{ or } x = -3.$$

4. Use the substitution for part (b)

The equation is the same as $f(u) = 0$ with $u = y^{2/3}$.

5. Choose the valid root

Since $y > 0$, $u = y^{2/3} > 0$. The valid root is $u = \frac{5}{2}$, so $y^{2/3} = \frac{5}{2}$.

6. Write y in surd form

$$y = \left(\frac{5}{2}\right)^{3/2} = \sqrt{\frac{125}{8}} = \frac{5\sqrt{10}}{4}.$$

Final answer

$$(a) x = -3, 0, \frac{5}{2}.$$

$$(b) y = \frac{5\sqrt{10}}{4}.$$

Question 5

Integration

5.

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

The curve C has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = \frac{12}{\sqrt{x}} + \frac{x}{3} - 4$
- the point $P(9, 8)$ lies on C

(a) find, in simplest form, $f(x)$

(5)

The line l is the normal to C at P

(b) Find the coordinates of the point at which l crosses the y -axis.

(4)

(Total for Question 5 is 9 marks)

Worked Solution - Question 5

1. Integrate $f'(x)$

$$f'(x) = 12x^{-1/2} + \frac{x}{3} - 4, \text{ so } f(x) = 24\sqrt{x} + \frac{x^2}{6} - 4x + c.$$

2. Use $P(9, 8)$

$$8 = 24\sqrt{9} + \frac{9^2}{6} - 4(9) + c = 72 + \frac{81}{6} - 36 + c.$$

3. Find c

$$72 + \frac{81}{6} - 36 = \frac{99}{2}, \text{ so } c = 8 - \frac{99}{2} = -\frac{83}{2}.$$

4. Find the tangent gradient at P

$$f'(9) = \frac{12}{3} + \frac{9}{3} - 4 = 3.$$

5. Find the normal gradient

The normal gradient is the negative reciprocal of 3, so it is $-\frac{1}{3}$.

6. Find the y -intercept

The normal through $(9, 8)$ is $y - 8 = -\frac{1}{3}(x - 9)$, so $y = -\frac{1}{3}x + 11$. It crosses the y -axis at $(0, 11)$.

Final answer

$$(a) f(x) = 24\sqrt{x} + \frac{x^2}{6} - 4x - \frac{83}{2}.$$

$$(b) (0, 11).$$

Question 6

Graphs of Functions

6. (a) Given that k is a positive constant such that $0 < k < 4$ sketch, on **separate axes**, the graphs of

(i) $y = (2x - k)(x + 4)^2$

(ii) $y = \frac{k}{x^2}$

showing the coordinates of any points where the graphs cross or meet the coordinate axes, leaving coordinates in terms of k , where appropriate.

(5)

- (b) State, with a reason, the number of roots of the equation

$$(2x - k)(x + 4)^2 = \frac{k}{x^2}$$

(1)

(Total for Question 6 is 6 marks)

Worked Solution - Question 6

Topic group

1. Key points for $y = (2x-k)(x+4)^2$

The graph has x-intercepts $(-4, 0)$ and $(\frac{k}{2}, 0)$. The root at $x = -4$ is a repeated root, so the graph touches the x-axis there.

2. Use $0 < k < 4$

Since $0 < k < 4$

3. Key features for $y = k/x^2$

The graph is positive on both sides of the y-axis, with asymptotes $x = 0$ and $y = 0$. It has no intercepts with the coordinate axes.

4. Compare the two graphs

The equation $(2x - k)(x + 4)^2 = \frac{k}{x^2}$ represents intersections between the two graphs.

5. State the number of roots

From the sketches, the positive branch of $\frac{k}{x^2}$ meets the increasing positive branch of $(2x - k)(x + 4)^2$ once only. Therefore there is one root.

Final answer

The graphs intersect once, so the equation has one root.

Question 7

Transformations

7.

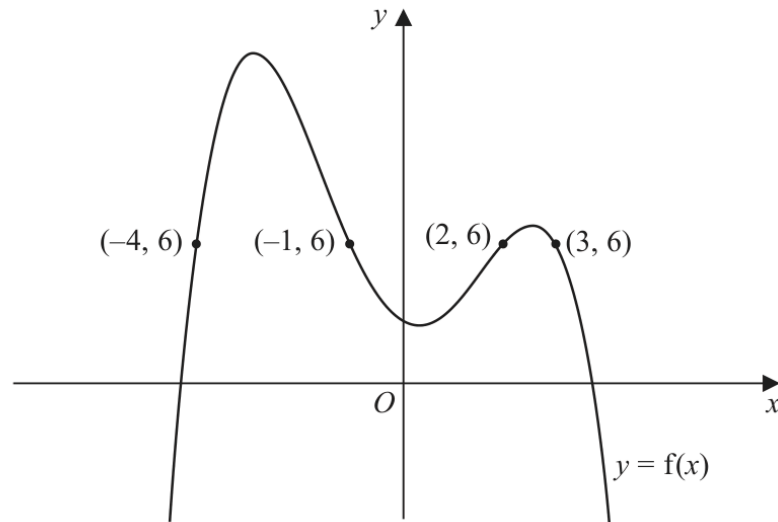


Figure 1

Figure 1 shows the curve with equation $y = f(x)$.

The points $P(-4, 6)$, $Q(-1, 6)$, $R(2, 6)$ and $S(3, 6)$ lie on the curve.

(a) Using Figure 1, find the range of values of x for which

$$f(x) < 6 \quad (3)$$

(b) State the largest solution of the equation

$$f(2x) = 6 \quad (1)$$

(c) (i) Sketch the curve with equation $y = f(-x)$.

On your sketch, state the coordinates of the points to which P , Q , R and S are transformed.

(ii) Hence find the set of values of x for which

$$f(-x) \geq 6 \text{ and } x < 0 \quad (4)$$

(Total for Question 7 is 8 marks)

Worked Solution - Question 7

1. Read $f(x) < 6$ from the graph

The curve is below the line $y = 6$ for $x < -4$, for $-13 < x < -4$.

2. Solve $f(2x) = 6$

Since $f(u) = 6$ at $u = -4, -1, 2, 3$, we set $2x = -4, -1, 2, 3$. The solutions are $x = -2, -\frac{1}{2}, 1, \frac{3}{2}$.

3. State the largest solution

The largest solution is $x = \frac{3}{2}$.

4. Reflect the points in the y-axis

For $y = f(-x)$, each point (x, y) becomes $(-x, y)$.

5. List the transformed points

$P(-4, 6) \rightarrow (4, 6)$, $Q(-1, 6) \rightarrow (1, 6)$, $R(2, 6) \rightarrow (-2, 6)$, and $S(3, 6) \rightarrow (-3, 6)$.

6. Use the reflected graph

For $x < 0$, the part of the reflected graph on or above $y = 6$ is from $x = -3$ to $x = -2$.

Final answer

(a) $x < -4$, $-13 < x < -4$.

(b) $\frac{3}{2}$.

(c)(ii) $-3 \leq x \leq -2$.

Question 8

Radians

8.

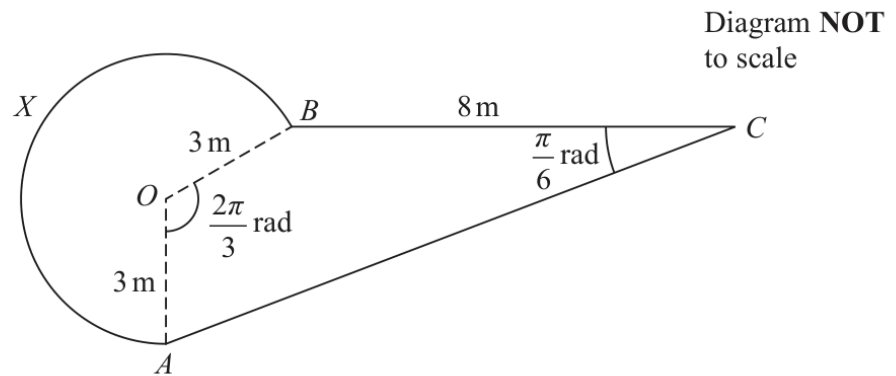


Figure 2

Figure 2 shows the plan view of a design for a pond.

The design consists of a sector $AOBX$ of a circle centre O joined to a quadrilateral $AOBC$.

- $BC = 8$ m
- $OA = OB = 3$ m
- angle AOB is $\frac{2\pi}{3}$ radians
- angle BCA is $\frac{\pi}{6}$ radians

(a) Calculate (i) the exact area of the sector $AOBX$,

(ii) the exact perimeter of the sector $AOBX$.

(5)

(b) Calculate the exact area of the triangle AOB .

(2)

(c) Show that the length AB is $3\sqrt{3}$ m.

(2)

(d) Find the total surface area of the pond. Give your answer in m^2 correct to 2 significant figures.

(5)

(Total for Question 8 is 14 marks)

Worked Solution - Question 8

1. Find the angle of the sector AOBX

The sector AOBX is the major sector, so its angle is $2\pi - \frac{2\pi}{3} = \frac{4\pi}{3}$.

2. Find the sector area

$$\text{Area} = \frac{1}{2}r^2\theta = \frac{1}{2}(3^2)\left(\frac{4\pi}{3}\right) = 6\pi \text{ m}^2.$$

3. Find the sector perimeter

The arc length is $r\theta = 3\left(\frac{4\pi}{3}\right) = 4\pi$. Adding the two radii gives $4\pi + 6 \text{ m}$.

4. Find the area of triangle AOB

$$\text{Area} = \frac{1}{2}(3)(3)\sin\left(\frac{2\pi}{3}\right) = \frac{9}{2} \cdot \frac{\sqrt{3}}{2} = \frac{9\sqrt{3}}{4} \text{ m}^2.$$

5. Show AB

By the cosine rule, $AB^2 = 3^2 + 3^2 - 2(3)(3)\cos\left(\frac{2\pi}{3}\right) = 18 + 9 = 27$, so $AB = 3\sqrt{3}$.

6. Find AC

In triangle ABC, $AB^2 = AC^2 + 8^2 - 2(AC)(8)\cos\left(\frac{\pi}{6}\right)$, so $27 = AC^2 - 8\sqrt{3}AC + 64$.

7. Choose the correct length

Solving gives $AC = 4\sqrt{3} \pm \sqrt{11}$. From the diagram, AC is the longer value, so $AC = 4\sqrt{3} + \sqrt{11}$.

8. Find the total area

Area of triangle ABC is $\frac{1}{2}(8)(AC) \sin\left(\frac{\pi}{6}\right) = 2AC = 8\sqrt{3} + 2\sqrt{11}$. Total area is $6\pi + \frac{9\sqrt{3}}{4} + 8\sqrt{3} + 2\sqrt{11} = 43.236\dots$, so 43 m^2 to 2 significant figures.

Final answer

(a)(i) $6\pi \text{ m}^2$, (ii) $4\pi + 6 \text{ m}$.

(b) $\frac{9\sqrt{3}}{4} \text{ m}^2$.

(c) $AB = 3\sqrt{3} \text{ m}$.

(d) 43 m^2 .

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14 marks

Question 9

Differentiation

9.

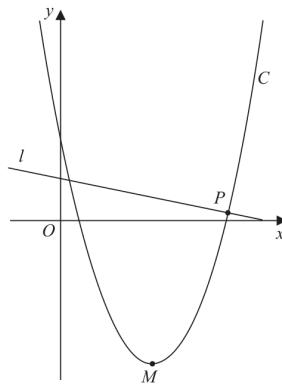


Figure 3

Figure 3 shows a sketch of the curve C with equation

$$y = \frac{1}{2}x^2 - 10x + 22$$

- (a) Write $\frac{1}{2}x^2 - 10x + 22$ in the form

$$a(x + b)^2 + c$$

where a , b and c are constants to be found.

(3)

The point M is the minimum turning point of C , as shown in Figure 3.

- (b) Deduce the coordinates of M

(2)

The line l is the normal to C at the point P , as shown in Figure 3.

Given that l has equation $y = k - \frac{1}{8}x$, where k is a constant,

- (c) (i) find the coordinates of P

- (ii) find the value of k

(6)

Question 9 continues on the next page

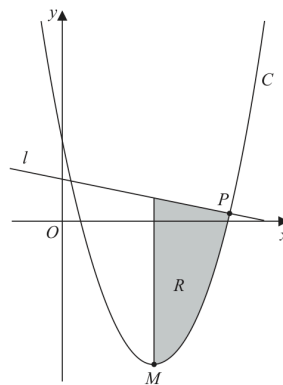


Figure 4

Figure 4 is a copy of Figure 3. The finite region R , shown shaded in Figure 4, is bounded by l , C and the line through M parallel to the y -axis.

- (d) Identify the inequalities that define R .

(3)

(Total for Question 9 is 14 marks)

Worked Solution - Question 9

1. Complete the square

$$\frac{1}{2}x^2 - 10x + 22 = \frac{1}{2}(x^2 - 20x) + 22 = \frac{1}{2}((x - 10)^2 - 100) + 22.$$

2. Simplify the completed square form

$$\frac{1}{2}((x - 10)^2 - 100) + 22 = \frac{1}{2}(x - 10)^2 - 28.$$

3. Read the minimum point

The minimum point is $M(10, -28)$.

4. Use the normal gradient

The line l has gradient $-\frac{1}{8}$, so the tangent gradient at P is 8.

5. Differentiate the curve

$$\frac{dy}{dx} = x - 10. \text{ At } P, x - 10 = 8, \text{ so } x = 18.$$

6. Find P

$$\text{When } x = 18, y = \frac{1}{2}(18)^2 - 10(18) + 22 = 4, \text{ so } P = (18, 4).$$

7. Find k

$$\text{Since } P \text{ lies on } y = k - \frac{1}{8}x, 4 = k - \frac{1}{8}(18) = k - \frac{9}{4}, \text{ so } k = \frac{25}{4}.$$

8. Write the upper boundary

$$\text{The line is } y = \frac{25}{4} - \frac{x}{8}.$$

9. Write the lower boundary

The curve is $y = \frac{1}{2}x^2 - 10x + 22$.

10. Identify the region

The region lies to the right of the vertical line through M and between the curve and the normal line: $10 \leq x \leq 18$ and $\frac{1}{2}x^2 - 10x + 22 \leq y \leq \frac{25}{4} - \frac{x}{8}$.

Final answer

(a) $\frac{1}{2}(x - 10)^2 - 28$.

(b) $M(10, -28)$.

(c) $P(18, 4)$, $k = \frac{25}{4}$.

(d) $10 \leq x \leq 18$, $\frac{1}{2}x^2 - 10x + 22 \leq y \leq \frac{25}{4} - \frac{x}{8}$.